February 2026

SOP 4
Water and Sediment eDNA



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PROTOCOL REFERENCE

California Molecular Methods Workgroup SOPs for DNA sampling best practices and water eDNA collection: https://github.com/stheroux/MMWG/

OBJECTIVE

Environmental DNA methods provide an opportunity to survey multiple taxa at one time from an environmental sample, including microbes, fish, and mammals. In conjunction with the other biological indicators, water eDNA samples will be collected and analyzed to profile species biodiversity. This SOP describes the collection of water and sediment eDNA samples.

We provide two eDNA filtration methods. The first method recommends filtering the field using the Smith Root Citizen Scientist Sampler. The second method provides details for water collection and filtering at a laboratory.

Field Methods: Smith-Root Citizen Scientist Sampler

Demonstration Video: https://youtu.be/BupozR1fnlg

SmithRoot citizen scientist manual: https://www.smith-root.com/support/downloads/edna-citizen-scientist-sampler-manual

Materials

1. SmithRoot 1.2 um self-preserving filters
2. SmithRoot citizen scientist sampler*
3. Labels
4. 500ml or 1L bottle (required if not sampling directly from water source)**
5. MilliQ or DI water
6. Nitrile gloves
7. DNAway or 10% bleach for cleaning

*Before heading to the field, ensure the sampler is fully charged. There is a charging cord included for the lead acid battery.

** Items should be sterilized before use and between sampling sites to prevent cross-contamination. To sterilize, soak in acid wash (1% solution of hydrochloric or nitric acid) or bleach (final concentration 1-5%), rinse 3x in DI H2O, and autoclave.

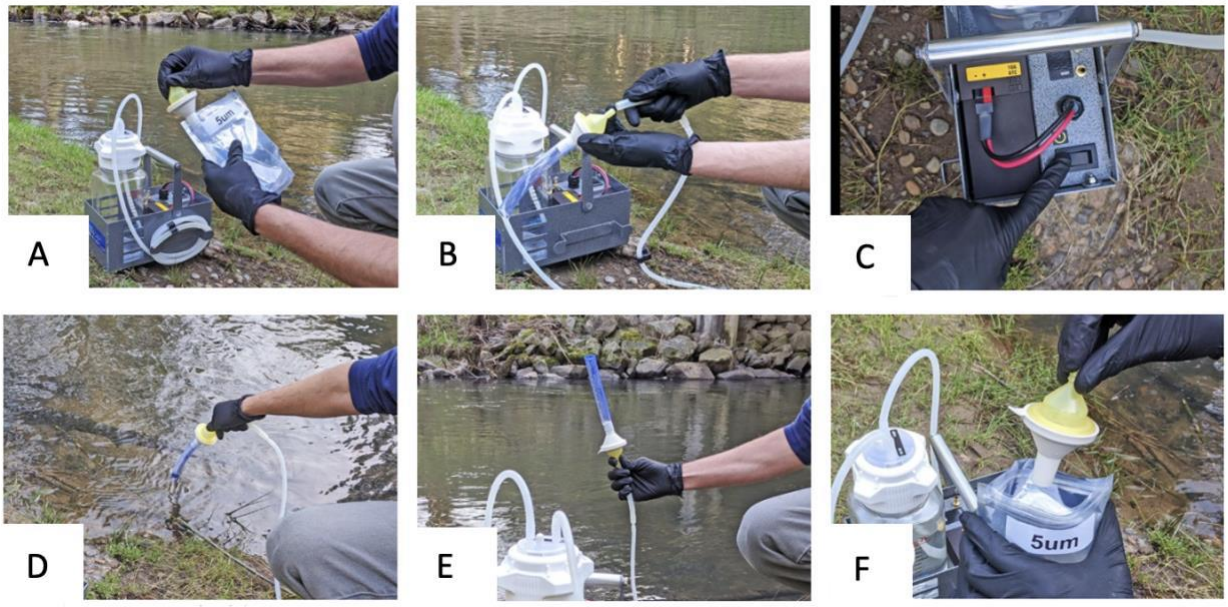


Figure 4.1 Overview of SmithRoot sampler filtration steps. Images from SmithRoot. June 2024.

Procedure

Sample collection

Three 1L Nalgene bottles will be used ***per sampling station*** – collecting 3 replicates. An additional 1L Nalgene bottle should be used to transfer 1L of MilliQ or DI water into a clean bottle. ***Only 1 field blank is needed per estuary.***

1. Rinse sample bottle 3x with ~10ml of sample water; discard rinse water offshore or away from sampling site. Fill bottle with 1L of sample water. If bottle numbers are limited, large WhirlPaks can be used for water collection instead. Skip this step if sampling directly from water source (e.g. river as in Figure 1D).

Filtering

1. Wearing gloves, open an individually-wrapped SmithRoot filter pouch. Without touching the snorkel tube, attach the snorkel tube to the smooth-end of the filter housing (**Figure 1A**).
2. Connect the Sampler hose to the ridged end of the filter (**Figure 1B**).
3. Turn on vacuum pump (**Figure 1C**) and insert snorkel tube into sample water (**Figure 1D**).
4. When 1L has been filtered, invert filter so that snorkel tube is facing the sky. Continue to run the vacuum pump for 20seconds to dry the filter (**Figure 1E**).

After filtering

1. Remove the snorkel tube from the filter housing and discard the snorkel.
2. Place the filter and filter housing back in the pouch (**Figure 1F**). Label pouch with Station, Date, Depth, Replicate, and Volume filtered. Record filter information on field data sheet.
3. Discard the filtrate in the collection vessel.
4. SmthRoot self-preserving filters are stable at room temperature for 6months.
5. Repeat all steps for as many replicates as requested (default = 3 replicates).

Blanks

1. Filter ~100ml of MilliQ water through a clean filter as above. Process one blank for each sampling day.

Cleaning

1. Seawater can corrode the metal pieces of the SmithRoot sampler. Please thoroughly wipe down unit after each field day with a wet paper towel. Store SmithRoot pump in clean, dry garbage bag when not in use.

Storage and Shipping

1. SmithRoot filters can be stored and shipped at room temperature for up to 6 months.
2. Ensure all samples are properly labelled according to the description above in the field methods
3. Properly box the samples for transport
4. Make a photocopy of the SOP4 field data sheet and sign the bottom. Include the copied form in the box to serve as a COC form for the lab.

FIELD METHODS: SAMPLE COLLECTION WITH LABORATORY FILTRATION

Field Materials

1. Nitrile gloves
2. 1L Nalgene bottles* labeled with sample site code, station number, date, and replicate number
3. Cooler
4. Ice

* Items should be sterilized before use and between sampling sites to prevent cross-contamination. To sterilize, soak in acid wash (1% solution of hydrochloric or nitric acid) or bleach (final concentration 1-5%), rinse 3x in DI H2O, and autoclave.

Field Procedure

Sample Collection

Three 1L Nalgene bottles will be used ***per sampling station*** – collecting 3 replicates. An additional 1L Nalgene bottle should be used to transfer 1L of DI water into a clean bottle. ***Only 1 field blank is needed per estuary.***

Samples should be kept **cold on ice** until back at the lab and then **frozen**.

Preparation

1. Prior to removing eDNA supplies or collecting the sample, put on **nitrile gloves.**

Collection of field blank

2. **Before sample collection**, one field blank should be processed per estuary.
    a. Pour 1L of DI water into a pre-labeled 1L Nalgene bottle.

Collection of water eDNA sample

3. Sample water should be collected at high tide using a 1L sample bottle. Surveyors can wade into the water, so that samples are collected in the intertidal zone at high tide. There should be at least 1 m of water at time of collection.
4. Stand in the water until the **sediment settles**, and then begin collecting the samples upstream of your location to avoid sampling where you've kicked up sediment.
5. Rinse sample bottle 3x with ~10ml of sample (estuarine) water; discard rinse water offshore or away from sampling site. Fill bottle with 1L of sample water. Leave headspace to allow for water expansion during freezing. If bottle numbers are limited, large WhirlPaks can be used for water collection.
6. Carry sample water back to shore.
7. After this stage**, samples should be brought back to the lab on ice.**
    a. Sample water should be filtered within **24 hours,** stored in the refrigerator until filtering within the 24-hour window
    b. Alternatively (not recommended) eDNA bottles could be frozen until ready for filtration.

Lab Materials

1. Filtration manifold, vacuum pump, vacuum tubing, vacuum flask*
2. Filters (0.45um cellulose nitrate, Millipore HAWP04700, Stericup, or otherwise specified)
3. 500ml or 1L sampling bottle*
4. MilliQ water
5. Nitrile gloves
6. Tweezers/forceps*
7. Whirl Pak bags labeled with sample site code, date, and replicate number

* Items should be sterilized before use and between sampling sites to prevent cross-contamination. To sterilize, soak in acid wash (1% solution of hydrochloric or nitric acid) or bleach (final concentration 1-5%), rinse 3x in DI H_2O, and autoclave. Rinse tweezers with DNAway followed by MilliQ between samples or flame sterilize with ethanol.

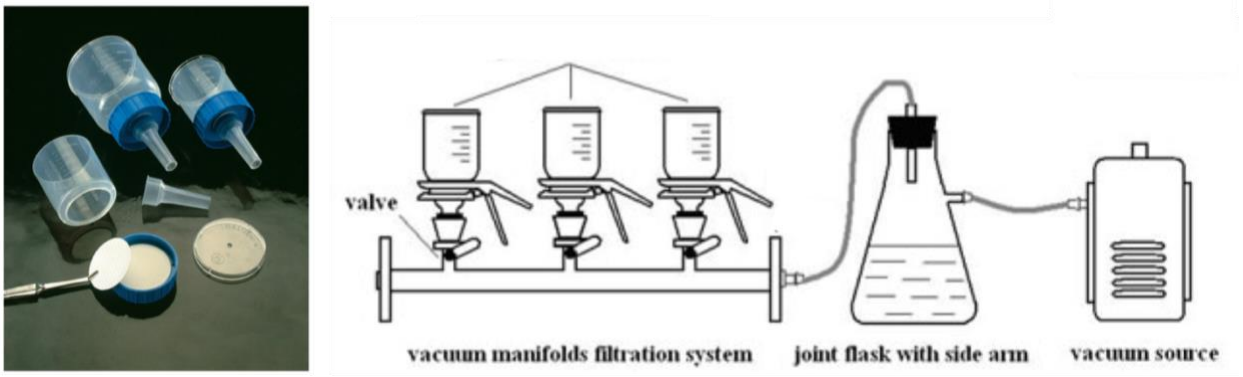


Figure 4.2. Left: Stericup filtration units. Right: vacuum manifold filtration set-up.

Lab Filtration

Prior To Processing

- Always wear gloves.
- Wipe down workspace with 10% bleach or DNAway.
- Be mindful of any surfaces you touch as door handles, counter tops, refrigerator/freezer door can contaminate your gloves (and you can contaminate surfaces). Change gloves frequently.

Filtering with filter funnel:

1. Place clean filter in filter holder filter device on vacuum manifold.
2. Vigorously shake bottle with sample to homogenize. Pour sample into filter cup and replace cover. Turn on vacuum pump and filter.
3. If the filter becomes clogged, make note of total volume of sample filtered. **Record this volume on worksheet.**

After filtering:

4. When filtration is complete, let the pump run 5-10 more seconds to dry filter, then turn off the pump.
5. To remove filter from the manifold, remove the filter cup and gently fold filter using clean forceps, optimally folding three times to create triangle-shaped filter with filtrate protected on the inside of the triangle. Place filter in pre-labeled 2ml screw cap tubes and cap.
6. Place labelled tube in labeled Whirl Pak bag.
7. Keep tubes with filters on ice in dark until able to be transported to -20°C or -80°C freezer.
8. Repeat all steps for as many replicates as requested (default = 3 replicates + 1 Field Blank).

Lab Blanks:

9. Filter ~100ml of MilliQ water through a clean filter and preserve as with field samples. Process one lab blank for each lab day.

FIELD METHODS: SEDIMENT

Sediment collection for eDNA should be paired with other sediment collections (SOP 3).

Materials

1. Nitrile gloves
2. 1.5mL Eppendorf tubes
3. Mini scraper
4. Cooler
5. Ice

Procedure

Sample Collection

Three sediment samples will be collected ***per sampling station*** – collecting 3 replicates. No sediment field blank is collected.

Samples should be kept **cold on ice** until back at the lab and then **frozen**.

Preparation

1. Prior to removing eDNA supplies or collecting the sample, put on **nitrile gloves.**

Collection of sediment eDNA sample

2. Sample sediment should be collected in the intertidal zone using a 1.5mL Eppendorf tube or similar.
    a. Intertidal: the channel edge and/or at the edge of the marsh plain, where the vegetation starts.
3. Before beginning, the mini scoop or scraper should be washed with seawater to remove residual sediment and rinsed with de-ionized (DI) water between samples.
4. At each collection spot, identify a ~5 x 5 cm, undisturbed area without vegetation
5. Wearing gloves, scrap the scoop or scraper along the top of the sediment and then scoop the sediment into the tube.
    a. Avoid touching the inside of the tube.
    b. Remove sediment along the groves of the tube, so that the lid attaches securely.
6. Tubes should be labeled as triplicates in order to pair location with replicate.
    a. For example, eDNA tubes from UCLA are labeled with the following nomenclature

   i. K-1045-B2-@, K-1045-B2-#, K-1045-B2-&
   ii. B2 represents a single location, and @, #, and & represent the 3 replicates
b. Record the sample code on the field datasheet and Estuary Checklist

7. **Samples should be brought back to lab on ice.**

Ship water eDNA sample

8. Sediment eDNA tubes should be frozen and shipped on dry ice to UCLA or designated laboratory.

QUALITY ASSURANCE/ QUALITY CONTROL (QA/QC)

Field personnel must be thoroughly trained in the proper use of sample collection gear and must be able to distinguish acceptable versus unacceptable water grab samples in accordance with pre-established criteria.

Field personnel must be thoroughly trained to recognize and avoid potential sources of sample contamination (e.g., non-sterile sampling supplies, poor sterile technique).

All data collected on the eDNA data sheet should be entered clearly. Team Leaders should double check that all eDNA samples have been collected and properly filtered and that the Data sheets have been filled out before leaving a transect site. eDNA samples relinquished to labs need to have the following eDNA COC documentation: SCCWRP eDNA CoC template_forBight.xlsx